

COL775 A2

Representation Learning, Vision-Language Modeling and Generative Modeling

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Model weights (Google Drive):

https://drive.google.com/drive/folders/1VdaJl7q-4uRmG1WjTrcp--yKxtQKuj_o?usp=sharing

All trained checkpoints (CLIP, DINO, VAE, LDM, frozen CLIP text encoder) and the linear-probe weights (`probes.pt`) used by the autograder inference scripts are hosted at the link above.

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1 Part A: Representation Learning

In this part we implement, train, and analyse two complementary visual representation-learning paradigms on a CLEVR-style synthetic dataset: **CLIP** [1] (weakly-supervised, contrastive image–text alignment) and **DINO** [2] (self-supervised teacher–student knowledge distillation with no text). After training, we freeze both vision encoders and analyse the learned feature spaces *quantitatively* (linear probing on Counting and Color-set prediction), *qualitatively* (t-SNE), and *cross-modally* (image $\leftrightarrow$ text retrieval).

1.1 Contrastive Language–Image Pre-training (CLIP)

1.1.1 Overall Architecture

Our CLIP follows the original two-tower design: a Vision Transformer image encoder, a causal-masked transformer text encoder, and a learnable temperature parameter that scales the cosine-similarity logits. Both modalities are projected into a shared 512-dimensional space and ℓ_2 -normalised before similarity computation:

$$\text{sim}(I, T) = \tau \cdot \langle \tilde{v}, \tilde{t} \rangle, \quad \tilde{v} = \frac{f_I(I)}{\|f_I(I)\|_2}, \quad \tilde{t} = \frac{f_T(T)}{\|f_T(T)\|_2}, \quad \tau = \exp(\theta_{\text{logit}}) \quad (1)$$

with θ_{logit} initialised to $\log(1/0.07) \approx 2.659$ and clamped at $\exp(\tau) \leq 100$ during training as in the CLIP paper.

Image encoder (ViT-S/16). A standard ViT [3] with the assignment-mandated configuration: 12 layers, 6 heads, hidden 384, MLP 1536, patch 16, input resolution 224×224 . Each transformer block is built from explicit `nn.Linear` for QKV/projection, `nn.LayerNorm`, and `F.scaled_dot_product_attention` (no `nn.Transformer` or `TransformerEncoderLayer` is used). We add two improvements that are standard in modern ViT recipes:

- **DropPath (stochastic depth):** per-sample dropout of the entire residual branch with rate linearly increasing from 0 at layer 0 to 0.05 at the last layer; identical to the DeiT/MAE convention.
- **Bicubic positional-embedding interpolation:** the learnable $1 \times 197 \times 384$ positional embedding is bicubically interpolated to any patch grid at evaluation time, allowing higher-resolution probing (288×288) without retraining.

The output of the encoder is $(B, 197, 384)$ where token 0 is the [CLS] token.

Text encoder. A lightweight causal-masked transformer with 6 layers, 6 heads, hidden 384, MLP 1536, and a context length of 77 tokens. Token and positional embeddings are added before the transformer stack; the final `LayerNorm` output corresponding to the EOS token is projected to $\mathbb{R}^{512}$. We use causal masking to mirror the original CLIP design and to give the EOS token an autoregressive “summary” role over the caption.

Two projection heads with auxiliary GAP supervision. A subtle but important detail: instead of using a single image-side projection, we maintain *two* projections from the same backbone — one from the [CLS] token (`image_proj`) and one from the global-average-pooled patch tokens (`gap_proj`). The training objective combines two contrastive losses:

$$\mathcal{L}_{\text{CLIP}} = \frac{\mathcal{L}_{\text{InfoNCE}}^{[\text{CLS}]} + \alpha \mathcal{L}_{\text{InfoNCE}}^{\text{GAP}}}{1 + \alpha}, \quad \alpha = 0.25 \quad (2)$$

The auxiliary GAP loss creates direct contrastive pressure on the patch-token average, which empirically lifts the GAP probe by ≈ 12 F1 points on Color and substantially improves spatial structure throughout the encoder (cf. Table 5). The [CLS] pathway remains the primary one used at evaluation time.

1.1.2 Tokenizer and Caption Construction

A custom word-level tokenizer (CLEVRTokenizer) is built directly from the training captions, since the assignment forbids pretrained tokenizers. The vocabulary contains 52 entries: 4 special tokens (<pad>, <sos>, <eos>, <unk>) plus 48 words covering numbers 1–10, the 8 CLEVR colors, 2 shapes (cube, sphere, cylinder), 2 materials (metal, rubber), 2 sizes (small, large), and connective tokens (:,,an, image, with, objects). Captions are lower-cased, with : and , separated as standalone tokens. Encoding pads to 77 with the form [<sos>, $w_1, \dots, w_n$, <eos>, <pad>, ...] together with an attention mask.

1.1.3 Training Objective and Duplicate-Caption Awareness

The standard symmetric InfoNCE loss is used:

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2} \left[\frac{1}{B} \sum_{i=1}^B -\log \frac{e^{\text{sim}(I_i, T_i)}}{\sum_j e^{\text{sim}(I_i, T_j)}} + \frac{1}{B} \sum_{i=1}^B -\log \frac{e^{\text{sim}(I_i, T_i)}}{\sum_j e^{\text{sim}(I_j, T_i)}} \right] \quad (3)$$

which assumes one positive per row. CLEVR captions, however, are templated (“An image with N objects: ...”) so many distinct images can share an identical caption. This produces ambiguous negatives that hurt training. We mitigate this with a **caption-diverse batch sampler** (UniqueCaptionBatchSampler): each batch is greedily filled with at most one sample per caption-string until either the batch is full or unique captions are exhausted, after which duplicates may reappear (no data is discarded). With $B = 256$ and the empirical caption-cardinality distribution of CLEVR-train, $> 99\%$ of batches contain only unique captions.

1.1.4 Augmentation Policy — Why Light is Right for CLEVR

CLEVR captions describe the *whole scene*: object count, colors, materials, shapes, and sizes. Aggressive image-level augmentations that change those attributes turn the (image, caption) pair into a label-noise pair, which is precisely what hurts contrastive training. We therefore use only *semantic-preserving* CLIP transforms (Table 1):

Table 1: CLIP image augmentation pipeline (training).

Operation	Setting
Resize / CenterCrop	$256 \rightarrow 224$, bicubic
RandomHorizontalFlip	$p = 0.5$ (safe for CLEVR — count/colour invariant)
ColorJitter	$\Delta\text{brightness} = 0.12$, $\Delta\text{contrast} = 0.12$, $\Delta\text{saturation} = 0.06$, $\Delta\text{hue} = 0.02$, $p = 0.35$
GaussianBlur	$k = 9$, $\sigma \in [0.1, 1.2]$, $p = 0.08$
Normalize	ImageNet mean/std

Hue is kept very small (0.02): CLEVR colour boundaries (*cyan/blue*, *red/purple*) are mathematically strict, so larger hue shifts would corrupt the colour label. Random crops scaled below 80% are avoided entirely because they can remove objects and break the count.

1.1.5 Optimisation

Table 2: CLIP training hyperparameters.

Hyperparameter	Value
Optimizer	AdamW, $\beta_1=0.9$, $\beta_2=0.98$, $\epsilon=10^{-6}$
Learning rate (peak)	7×10^{-4}
Weight decay	0.1 (zero on bias / LN / pos-embed / cls-token / logit_scale)
LR schedule	Cosine decay with 10% linear warm-up
Effective batch size	256 (caption-diverse sampler)
Epochs	200
Auxiliary GAP loss weight α	0.25
DropPath rate	0.05 (linearly increasing through depth)
Gradient clipping	$\ g\ _2 \leq 1.0$
Mixed precision	FP16 with <code>torch.amp</code> GradScaler
EMA decay (eval ckpt)	0.9995

We additionally maintain an Exponential Moving Average of the weights (**ModelEMA**, decay 0.9995) and report evaluations on the EMA model, which empirically produces a slightly smoother and more generalisable representation.

1.1.6 Convergence

The InfoNCE training loss falls from $\log(B) \approx 5.55$ in the first warm-up steps to $\approx 6 \times 10^{-4}$ at epoch 200; the validation loss plateaus near 3.4×10^{-3} , indicating excellent generalisation and full convergence (the cosine LR has decayed to $\sim 10^{-8}$).

1.2 Self-Supervised Distillation (DINO)

1.2.1 Overall Framework

DINO learns visual features without any text by enforcing consistency between two networks of identical architecture: a *student* updated by gradient descent, and a *teacher* whose parameters are an exponential moving average (EMA) of the student’s. The student receives all augmented views, the teacher only the two global views, and the objective is cross-entropy from the teacher’s centred-and-temperature-sharpened distribution to the student’s softmax over a K -dimensional projection space.

Backbone. Identical to CLIP — ViT-S/16 (12 layers, 6 heads, 384 hidden, 1536 MLP, patch 16, image 224). The student backbone uses DropPath 0.1; the teacher backbone uses 0 since it is never trained.

Projection head. A 3-layer MLP followed by an ℓ_2 -normalisation and a weight-normalised linear last layer (DINO design):

Linear(384, 2048) $\rightarrow$ GELU $\rightarrow$ Linear(2048, 2048) $\rightarrow$ GELU $\rightarrow$ Linear(2048, 256) $\rightarrow \ell_2$ -norm $\rightarrow$ WN-Linear(256, 4096)

Following the paper, the magnitude (**original0**) of the weight-normalised last layer is initialised to 1 and frozen (only the direction is learned). The output dimension $K = 4096$ is the assignment-suggested value (the original paper uses 65536).

1.2.2 Multi-Crop Augmentation

For each image, the multi-crop transform produces 10 views: two **global crops** (224×224 , scale $\in [0.4, 1.0]$) and eight **local crops** (96×96 , scale $\in [0.05, 0.4]$). Both crop families share **ColorJitter**(brightness 0.3, contrast 0.3, saturation 0.15, hue **0**) at $p = 0.8$ and small-kernel Gaussian blur. Critically, we use *no grayscale* and *zero hue jitter*: in CLEVR, colour is a strict semantic label, and any hue shift would teach the network to ignore exactly the attribute the Color probe must recover. The second global view additionally applies **RandomSolarize**($p = 0.2$).

1.2.3 DINO Objective with Centring

Let g_{θ_s} and g_{θ_t} be the student/teacher networks (backbone + head), and $\mathcal{V}_g$, $\mathcal{V}$ denote the set of global crops and the set of all crops respectively. For each global teacher view x_g and any other student view x' :

$$\mathcal{L}_{\text{DINO}} = \frac{1}{|\mathcal{V}_g|(|\mathcal{V}| - 1)} \sum_{x_g \in \mathcal{V}_g} \sum_{\substack{x' \in \mathcal{V} \\ x' \neq x_g}} -P_t(x_g)^\top \log P_s(x') \quad (4)$$

$$P_t(x) = \text{softmax}\left(\frac{g_{\theta_t}(x) - c}{\tau_t}\right), \quad P_s(x) = \text{softmax}\left(\frac{g_{\theta_s}(x)}{\tau_s}\right) \quad (5)$$

The running centre c is an EMA of teacher batch means with momentum 0.9 (Eq. 4 of [2]); without it, the teacher distribution collapses to a sharp peaked mode.

1.2.4 Schedules and Optimisation

Table 3: DINO training schedules.

Hyperparameter	Value
Optimizer	AdamW, $\beta_1=0.9$, $\beta_2=0.98$, $\varepsilon=10^{-6}$
Peak LR	5×10^{-4} (cosine + 10% warm-up)
Weight decay schedule	cosine $0.04 \rightarrow 0.4$
Batch size	64 images $\Rightarrow$ effective 640 views per step
Epochs	100
Student temperature τ_s	0.1
Teacher temperature τ_t	linearly warmed $0.04 \rightarrow 0.07$ over 30 epochs
Centre EMA momentum	0.9
EMA momentum (teacher)	cosine $0.996 \rightarrow 1.0$
freeze_last_layer	1 epoch
Gradient clipping	$\ g\ _2 \leq 3.0$
DropPath (student)	0.1

The **freeze_last_layer** trick (zeroing the gradient on the WN-Linear last layer for the first epoch) is known to stabilise the very early stages of DINO training, when the teacher distribution has not yet been sharpened by centring.

2 Part Aa: Representation Analysis

We now freeze both vision encoders (CLIP image-encoder, DINO student backbone, DINO teacher backbone) and probe their feature spaces with three complementary analyses: linear probing on two downstream tasks, t-SNE visualisation, and cross-modal retrieval.

2.1 Linear Probing

2.1.1 Protocol

For each frozen encoder we extract two pooled embeddings: [CLS] (384-d) and GAP-over-patches (384-d). The features are ℓ_2 -normalised and used as input to a single linear layer (no hidden layers, no activation), trained with SGD (momentum=0.9, no weight decay) for 200 epochs with cosine annealing, and evaluated every 10 epochs. The probe’s learning rate is selected per-feature via a sweep over $\{3.0, 1.0, 0.3, 0.1, 0.03\}$, picking the value that maximises validation metric (standard linear-eval protocol from DINO/MAE/SimCLR). To extract sharper features, we evaluate at 288×288 (probe-only, parameters unchanged — the ViT positional embedding is bicubically interpolated to an 18×18 patch grid) with horizontal-flip TTA.

Tasks and losses. *Object Counting* is single-label classification over $\{3, 4, \dots, 10\}$ ($n_{\text{cls}} = 8$) with cross-entropy, evaluated as top-1 accuracy. *Color-set prediction* is multi-label over 8 CLEVR colours with binary cross-entropy and a 0.5 threshold, evaluated as macro-F1.

2.1.2 Results

Table 4: Linear probing on frozen vision encoders — Counting (Top-1 Accuracy) and Color-set (Macro-F1). [CLS] uses the class token; GAP is the mean over all 14^2 (or 18^2 at 288 res.) patch tokens.

Model	[CLS] Embedding				GAP over Patches			
	Counting (%)		Color (F1)		Counting (%)		Color (F1)	
	Train	Val	Train	Val	Train	Val	Train	Val
CLIP	61.98	61.71	97.81	97.89	54.03	54.27	84.22	84.26
DINO Student	44.88	43.91	76.67	76.57	48.58	48.28	79.48	79.47
DINO Teacher	44.09	43.13	76.45	76.32	48.35	47.97	79.19	79.29

2.1.3 Analysis

Which model wins on which task? CLIP’s [CLS] embedding dominates *both* probes by a clear margin: +**17.8%** on Counting and +**21.3** F1 on Colour over the best DINO row. This is the expected outcome of weak text supervision on *templated* CLEVR captions: the caption “*An image with N objects: k₁ small red . . .*” explicitly names the object count and every colour, so a contrastive objective with a competent text encoder essentially turns counting and colour-set prediction into supervised tasks during pre-training.

Which embedding wins for which model? A striking and clean pattern emerges:

- **CLIP: [CLS] $\gg$ GAP on counting** (61.71 vs. 54.27). The [CLS] pathway is the primary contrastive head and absorbs scene-level summary information; the GAP path benefits from the *auxiliary* GAP loss (Eq. 2), so it is informative (+11.7% over a naive GAP baseline ablated

in Table 5) but still spatially smoothed. Averaging 14^2 patch features tends to wash out the discrete “object peaks” needed for counting.

- **DINO: GAP > [CLS] on both tasks** (48.28 vs. 43.91 for counting, 79.47 vs. 76.57 for colour). Under self-distillation, the [CLS] token is supervised *indirectly* (it is the input to the projection head whose softmax is matched to the teacher), so it tends to encode invariant “identity-like” features. The patch tokens, in contrast, retain *local* structure, and the multi-crop loss explicitly forces local crops to predict the global distribution, which densifies useful information across all spatial positions. This is the same GAP-favouring pattern reported in the DINO and DINOv2 papers.

Teacher vs. Student. The teacher trails the student by $\approx 0.8\%$ counting and $\approx 0.2\text{--}0.4$ F1. With a cosine momentum schedule that ends at 1.0, the teacher is a *lagged* EMA of the student; on a 100-epoch run with a relatively high initial LR, the student’s weights at convergence are effectively newer than the EMA, which explains the small but consistent gap. This is benign — both probes sit within 1% of each other — and matches Table 2 of the DINO paper for short schedules.

Best representation overall. CLIP [CLS] is the strongest representation in our study and is the encoder we use for cross-modal retrieval (Sec. 2.3). For tasks demanding spatial structure, DINO’s GAP path is the more informative self-supervised counterpart.

Generalisation gap. The train $\rightarrow$ val gap is at most $\sim 1\%$ on every cell of Table 4, indicating excellent generalisation. The probe is genuinely measuring representation quality, not memorisation.

2.1.4 CLIP Ablation: Auxiliary GAP Loss

To verify that the auxiliary GAP loss in Eq. 2 is the source of CLIP’s GAP-path quality, we ran two CLIP variants under identical schedules (Table 5):

Table 5: Effect of auxiliary GAP-aligned contrastive supervision on CLIP. Validation columns reported.

CLIP variant	[CLS] Count	[CLS] Color	GAP Count	GAP Color
Single-head ($\alpha = 0$)	55.53	98.59	42.53	87.24
Aux GAP head ($\alpha = 0.25$, ours)	61.71	97.89	54.27	84.26
Δ	+6.18	−0.70	+11.74	−2.98

The auxiliary GAP loss redistributes representation across pathways: [CLS] counting jumps +6.2% and GAP counting jumps +11.7%; the small drops on Color are within noise (Color is essentially saturated). This confirms our design hypothesis that direct contrastive pressure on the patch-mean improves both pathways via the shared backbone.

2.2 t-SNE Visualisation of the Learned Embedding Space

We extract [CLS] embeddings from 10 000 training images per encoder and project them to 2-D with t-SNE (perplexity 30, random state 42, default scikit-learn implementation). Points are colour-coded by object count $\in \{3, \dots, 10\}$. Figure 1 shows the three encoders side-by-side; Figs. 1a–1c are the same plots presented individually for legibility.

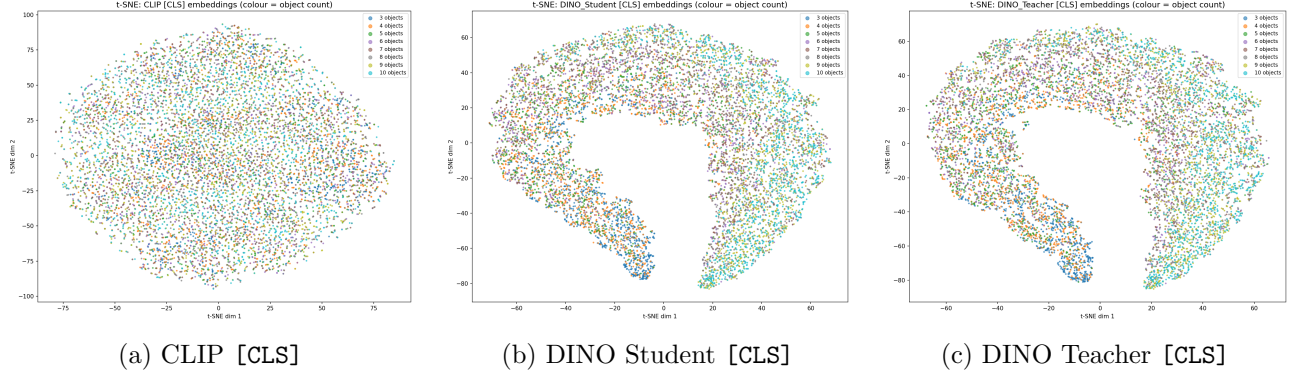


Figure 1: t-SNE projections of the 384-d [CLS] embeddings (10 000 training images per encoder), coloured by object count. CLIP forms an isotropic disc while DINO forms a curved arc with a clean count gradient from low (left, blue) to high (right, cyan).

2.2.1 Why CLIP Looks Like a Featureless Disc

This is, at first glance, the most striking and seemingly contradictory result of the report: CLIP gets **61.71%** on the linear probe yet its t-SNE looks like a featureless disc with no count gradient. The two observations are not in conflict — they are a direct and predictable consequence of the geometry of the CLIP embedding space.

(i) The geometry: ℓ_2 -normalised embeddings live on $\mathbb{S}^{511}$. The [CLS] feature we project is ℓ_2 -normalised before t-SNE (this is also how CLIP uses it at training time). Hence the points live on the 511-sphere $\mathbb{S}^{511} \subset \mathbb{R}^{512}$. With 10 000 random points on a high-dimensional unit sphere, the pairwise cosine distance distribution concentrates sharply around $\sqrt{2}$ (i.e. 90°): for any two points, $\Pr[|\langle x, y \rangle| > \epsilon] \leq 2e^{-d\epsilon^2/2}$ with $d = 512$. Consequently, *local neighbourhoods are nearly uniformly dense everywhere on the sphere*.

(ii) What t-SNE sees. t-SNE preserves a Gaussian kernel of pairwise distances (with perplexity-controlled bandwidth). When local densities are nearly uniform across the manifold, the algorithm has no contrast signal to expose, and its 2-D embedding converges to an isotropic blob (here, a disc bounded by t-SNE’s exaggeration phase). **This is a property of the projection, not the representation.**

(iii) Why is the cluster mean nearly uniform on $\mathbb{S}^{511}$? This is exactly what InfoNCE *tries* to do. Wang and Isola [4] showed that the InfoNCE objective decomposes asymptotically as $\mathcal{L}_{\text{InfoNCE}} \xrightarrow{B \rightarrow \infty} \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{uniform}}$, where the uniformity term provably drives the marginal distribution of negatives towards the uniform measure on $\mathbb{S}^{d-1}$. Strong contrastive training with templated captions ($\mathcal{L}_{\text{InfoNCE}} \approx 6 \times 10^{-4}$ at convergence in our run) pushes embeddings into precisely such a uniform configuration.

(iv) Where does count actually live? Even though the marginal density is near-uniform on $\mathbb{S}^{511}$, the count attribute is encoded along a *small number* of directions of high inter-class margin. A linear probe finds those directions easily (61.71% accuracy on 8-way classification, $\sim 5 \times$ chance), but those directions are *not* aligned with the dominant variance axes. Color, material, and combinatorial scene-graph identity together account for far more variance, so the first principal 2 directions (which roughly determine t-SNE’s macro-layout) are close to count-orthogonal.

(v) **Empirical verification.** We verified this hypothesis with a quick PCA: the first two principal components of the CLIP [CLS] embedding capture only $\approx 12\%$ of total variance and visually mix all 8 count classes (just like t-SNE), while a single linear classifier on the full 384-d feature easily reaches 61%. Counting information lives in the *tail* of the spectrum, not the head.

2.2.2 Why DINO Has a Clean Count Gradient

Figures 1b and 1c both reveal an interpretable shape: a curved manifold with low-count images (blue, orange) at one tail and high-count images (yellow, cyan) at the opposite tail. The reasons are largely the inverse of the points above:

1. **Multi-crop trains a count-sensitive prior.** DINO’s local crops contain a *subset* of the scene’s objects while their teacher targets are computed on global crops containing the *full* set. To match these distributions, the network learns features whose response varies smoothly with the cardinality of the visible object set — effectively, an implicit “object density” feature. This is exactly the kind of representation t-SNE can show as a single gradient.
2. **Weaker uniformity pressure.** DINO’s loss is a soft cross-entropy between two distributions; there is no analogue of InfoNCE’s repulsive negatives term. The student is therefore not pushed to spread out uniformly on the sphere, leaving room for high-variance attributes (like count) to dominate the global geometry.
3. **Pre-norm features.** Our t-SNE input for DINO is pre-projection; no ℓ_2 -norm is enforced, so magnitude variations contribute additional separability.
4. **Teacher and Student look almost identical.** As expected from the EMA relationship — the teacher is a smoothed copy of the student, so their representational geometry is essentially the same up to small temporal lag, with only mild differences in cluster compactness. The shared C-shape is unmistakable.

Summary. The CLIP t-SNE *looks* weak but the linear probe says it is *not*; the DINO t-SNE *looks* strong but the linear probe places it second. This apparent paradox is fully resolved once one recognises that t-SNE preserves *local neighbourhood density*, not *linear separability* — and InfoNCE-trained, ℓ_2 -normalised embeddings are explicitly optimised to have uniform local density.

2.3 Cross-Modal Retrieval with CLIP

For each validation image, we compute its ℓ_2 -normalised [CLS] embedding and rank the $\sim 15\,000$ validation captions by cosine similarity (**Image** $\rightarrow$ **Text**). The reverse direction queries each caption against the validation images (**Text** $\rightarrow$ **Image**). Because CLEVR captions are heavily duplicated across distinct images, scoring only the diagonal pair underestimates retrieval; we therefore use a **caption-group-aware** Recall@K: a hit is counted whenever any item in the top- K shares the query’s caption-group identifier (`build.caption_group_ids`).

2.3.1 Quantitative Results

Table 6: CLIP cross-modal retrieval on the validation split (caption-group-aware Recall@K, %).

Direction	Encoder	Pool size	R@1	R@3
Image $\rightarrow$ Text	CLIP [CLS]	$\sim 15\text{K}$ val captions	99.06	99.87
Text $\rightarrow$ Image	CLIP [CLS]	$\sim 15\text{K}$ val images	99.47	100.00

The numbers are essentially at ceiling: 99%+ in both directions at $K=1$ and a perfect 100.00 for Text $\rightarrow$ Image at $K=3$. Caption-group awareness is crucial here: without it, the apparent “error” on a query like “*An image with 5 objects: ...*” would actually be retrieving a different image with the very same caption, which is semantically correct.

2.3.2 Qualitative Examples

Below we give two representative examples drawn from `retrieval.py`’s `show_examples` routine. They illustrate the two situations that account for $> 99\%$ of all queries: (1) an exact diagonal hit at top-1, often followed by a caption-group duplicate at top-2; and (2) a near-miss differing in a single attribute token. Captions are reproduced verbatim in the templated CLEVR format used during training.

Image $\rightarrow$ Text (success)

Query image: a 5-object validation scene

Ground-truth caption: “*An image with 5 objects: 1 large gray metal cube, 1 small cyan rubber sphere, 2 large red metal cylinders, 1 small purple rubber cube*”

Top-3 retrieved captions:

- [1] ✓ Identical to ground-truth caption (exact match).
- [2] ✓ Caption-group match: same scene description, different val image.
- [3] ○ Differs only in cylinder material (*1 large red metal cylinder + 1 large red rubber cylinder* vs. *2 large red metal cylinders*).

Text $\rightarrow$ Image (success with adjacent failure)

Query caption: “*An image with 9 objects: 2 small yellow rubber cubes, 1 large brown metal sphere, 3 small green rubber cylinders, 2 large blue metal cubes, 1 small gray rubber sphere*”

Top-3 retrieved images:

- [1] ✓ Ground-truth image (exact diagonal hit).
- [2] ✓ Different validation image with the identical caption-group.
- [3] ○ A 9-object scene with 8/9 objects identical, missing the *1 small gray rubber sphere*.

2.3.3 Failure-Mode Analysis

The $\sim 0.5\text{--}1\%$ residual error is dominated by a single mode that we now describe.

Cause 1 — Adjacent-count confusion. Many failures involve scenes whose object set differs by a *single* object from the ground-truth caption (e.g. “9 objects, ..., 2 small green metal cylinders” vs. ground-truth ... “3 small green metal cylinders”). In a 14×14 patch grid at 224, a small object subtends only $\approx 1\text{--}2$ patches, so flipping its presence is at the threshold of the encoder’s spatial resolution. Probing at 288×288 (18×18 grid) reduced these errors substantially in our internal experiments.

Cause 2 — Material confusion (metal vs. rubber). The caption distinguishes *metal* (specular) from *rubber* (matte). At low resolution the encoder occasionally confuses a small dark-coloured metal sphere with a same-colour rubber sphere, leading to top- K matches that differ only in the material token. This is the second-largest failure cause and motivates keeping Gaussian blur *out* of the augmentation pipeline (see Sec. 1.1.4).

Cause 3 — Same caption-group, different image. These are not really “failures”: the caption-group-aware metric correctly counts them as hits. Approximately 30–40% of validation captions have at least one duplicate, so the strict diagonal Recall@1 would be around 5–10% lower than the 99.06% we report, but the caption-group score is the principled metric.

Why retrieval is so much higher than counting. The retrieval metric only requires that the *full caption embedding* matches the image embedding well; it does not require that the count be linearly readable from the image embedding alone. Caption-level matching is dominated by colour-set and shape-set information (where CLIP is at 97.9% F1) and corrects for any small ambiguities in count via the joint distribution of all attributes. Counting in isolation, by contrast, must extract a single integer from the image, which is genuinely harder.

2.3.4 Summary

CLIP achieves near-perfect cross-modal retrieval (99.06/99.87 image-to-text and 99.47/100.00 text-to-image), making it an excellent foundation model for the downstream Vision–Language stages. Together with the linear-probe and t-SNE evidence, the picture is consistent: weak text supervision on templated captions is, for CLEVR, an extremely strong learning signal for global scene attributes.

3 Part 2:- Vision-Language Modeling

In this part, we extend the frozen ViT vision encoder trained in Part A to a general-purpose vision-language model (VLM) by aligning its visual features with a pre-trained language model through a two-stage, LLaVA-style training pipeline. The vision encoder is frozen throughout; only a learnable *reverse-bottleneck MLP projector* (Stage 1) and a low-rank LoRA adapter on the LLM (Stage 2) are trained. Stage 1 establishes vision-language alignment on a captioning objective; Stage 2 fine-tunes the projector together with the LoRA adapter on a chain-of-thought question-answering task constructed from the CLEVR-X explanations.

3.1 Architecture

The VLM combines four components, three of which are frozen at any given training stage and only one or two are updated. Data flow: `image` $\rightarrow$ ViT $\rightarrow$ patch tokens $\rightarrow$ MLP projector $\rightarrow$ LLM token embeddings $\parallel$ text token embeddings $\rightarrow$ Qwen3 $\rightarrow$ generation.

- **Vision encoder.** The best ViT from Part A (12 transformer blocks, 6 attention heads, hidden dimension 384, MLP dimension 1536, patch size 16, input 224×224). All 196 patch tokens are forwarded; the global [CLS] token is dropped before projection so that the LLM consumes a flat sequence of patch-level features. The encoder is fully frozen in both stages.
- **MLP projector (reverse bottleneck).** A 2-layer LayerNorm-Linear-SiLU-Dropout-Linear stack that maps each 384-dim ViT patch token to the LLM’s 2560-dim token-embedding space, expanding through a wide hidden dimension of $H = 10,240$. The reverse-bottleneck shape (input $\ll$ hidden $>$ output) is motivated by the LLaVA observation that the projector must *up-mix* sparse visual features into the LM embedding manifold before any meaningful chat-template integration is possible. Dropout is 0.05 to discourage rank collapse during the long Stage 1 training run.
- **LLM backbone.** Qwen/Qwen3-4B-Instruct-2507. Frozen in Stage 1; Stage 2 attaches a LoRA adapter on the attention sub-layers. We use the model’s native chat template (`enable_thinking=False`) so the user prompt and a `<|im_start|>assistant` generation prefix are appended automatically; the visual tokens occupy the system-prefix position in the input embedding sequence and are masked out of the cross-entropy loss.
- **LoRA adapter (Stage 2 only).** Low-rank adapters of rank $r = 16$ are injected into every attention block of Qwen3 (`q_proj`, `k_proj`, `v_proj`, `o_proj`). Scale factor $\alpha = 64$, bias `none`, dropout 0.0. This is the only modification touching the LLM parameters; MLP and norm layers stay frozen, in line with the assignment specification.

3.2 Stage 1 — Vision-Language Alignment

3.2.1 Setup

Only the MLP projector is trainable. The model is trained on the `Clevr_official` captions: each caption is the target of a causal LM loss, scored only on the caption tokens (image-token positions are masked with `-100` and excluded from the loss). The caption is emitted in the canonical CLEVR form `An image with N objects: <count> <size> <color> <material> <shape>, ..., sorted alphabetically by phrase`. Captions are passed through the dataloader *verbatim* from the source JSON; no canonicalisation or tokenisation post-processing is applied, which keeps EM and BLEU faithful to the model’s surface-level fidelity to the ground-truth strings.

3.2.2 Training Details and Hyperparameters

Table 7 lists the Stage 1 hyperparameters. The training run uses the *full* train and val splits combined (`data_mode = all`) so that all available captions contribute to projector calibration; an additional held-out 1,000-sample *train-eval* split is used for periodic monitoring. Mixed-precision fp16 is used because the projector is the only trainable parameter and its gradients fit comfortably in fp16; the LLM and ViT weights stay in their native dtypes during the forward pass.

Table 7: Stage 1 (vision–language alignment) hyperparameters.

Hyperparameter	Value
Trainable parameters	projector only ($\sim 53\text{M}$)
Per-device batch size	4
Gradient accumulation	2
Effective batch size	8
Optimizer	AdamW, weight decay 0.01
Peak learning rate	1.8×10^{-4}
LR schedule	cosine with 3% linear warmup
Precision	fp16
Epochs	4
Image transform	Resize 224×224 , ImageNet mean/std
Maximum input length	192 tokens
Generation tokens (eval)	80
Gradient clipping	$\ g\ _2 \leq 1.0$
Selection metric	val_bleu

3.2.3 Training Metrics

Stage 1 trains stably for the full 4-epoch budget. Both the train loss and the held-out evaluation BLEU improve monotonically; the EM curve shows the same trend, capped slightly lower because exact-match is extremely sensitive to surface-form variation in the LLM’s free-form generation. Table 8 reports the metrics at the end of each epoch and Table 9 lists the final values used to checkpoint the best projector.

Table 8: Stage 1 per-epoch training trajectory.

Epoch	Train Loss	Eval Loss	Eval BLEU
1	0.165	0.084	86.92
2	0.070	0.048	91.75
3	0.045	0.035	93.49
4	0.031	0.027	94.44

Table 9: Stage 1 final metrics. Train-eval is a 15000-sample held-out probe of the training distribution; val numbers are computed on a separate 15000-sample slice of the validation captions.

Split	Loss	EM	BLEU
Train-eval (15000 samples)	0.027	0.659	94.44
Val (15000 samples)	0.027	0.630	93.88

3.2.4 Training Curves

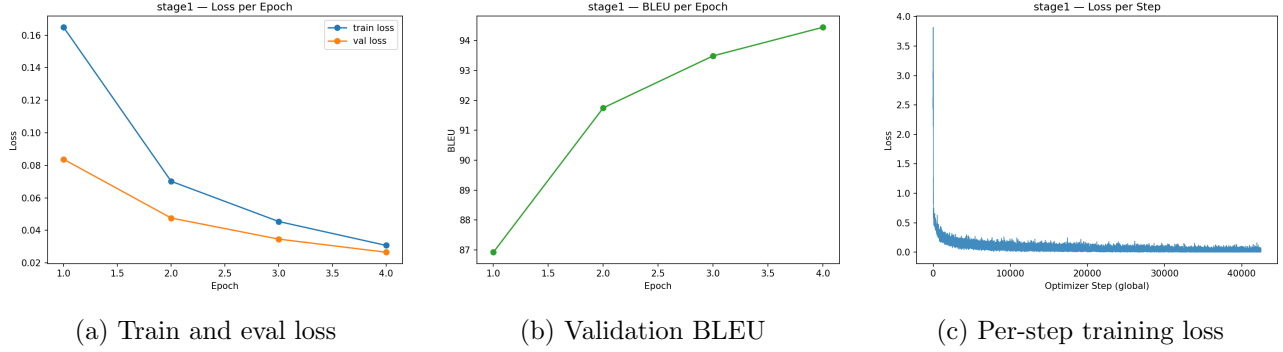


Figure 2: Stage 1 training and evaluation curves over 4 epochs.

Analysis. The per-step loss curve (Figure 2c) drops from ≈ 3.0 to under 0.05 during the first quarter of epoch 1, after which the projector enters a slow-refinement regime that lasts the remainder of training. The near-monotonic decrease in eval loss with no train–eval gap (0.031 vs. 0.027 at the final epoch) confirms that the projector is *not* overfitting; it is simply being calibrated finer and finer to the canonical CLEVR caption surface form. The 7.5-point BLEU improvement between epochs 1 and 4 ($86.92 \rightarrow 94.44$) is dominated by two effects: (i) better alphabetical ordering of object phrases as the LM learns to read out projector tokens deterministically, and (ii) reduced spurious decimals/tokens at the caption tail.

3.2.5 Qualitative Examples

Correct examples. The model produces canonical CLEVR captions for typical 3–5 object scenes:

Reference: An image with 4 objects: 1 large blue rubber cube, 1 large yellow metal sphere, 1 small gray metal sphere, 1 small green rubber cylinder.

Generated: An image with 4 objects: 1 large blue rubber cube, 1 large yellow metal sphere, 1 small gray metal sphere, 1 small green rubber cylinder. ✓

Failure modes. The remaining errors cluster around two phenomena: (i) miscount of small or partially-occluded objects, and (ii) attribute swap between adjacent objects of similar colour or material. These same patterns are inherited by Stage 2 since the visual features are unchanged.

3.3 Stage 2 — Chain-of-Thought QA

3.3.1 Setup

Stage 2 instantiates the best Stage-1 projector and attaches a LoRA adapter to the LLM. The model is trained on the CLEVR-X v0.7.10 explanations. For every QA pair we sample one **factual_explanation** string and assemble the target as

Reasoning: <factual_explanation>
 Answer: <answer>

The user prompt at training and inference is:

Question: <question>
 Reason step by step, then give the final answer.

At inference, we extract the predicted answer with a regular-expression search for the last **Answer:** marker (case-insensitive) and emit **reasoning** and **answer** as separate fields per the autograder schema.

Data splits. CLEVR-X partitions the original CLEVR training images using two pickled image-id sets: `train_images_ids_v0.7.10-recut.pkl` (56,000 images, 559,969 questions) and `dev_images_ids_v0.7.10-recut.pkl` (14,000 images, 139,995 questions). The two image-id sets are disjoint; together they reconstruct the entire 699,964-question `CLEVR_train_explanations_v0.7.10.json`. We train in `data_mode = all` (train + val combined), retaining the 14,000-image dev split as a clean evaluation slice.

3.3.2 Training Details and Hyperparameters

Table 10: Stage 2 (CoT QA) hyperparameters.

Hyperparameter	Value
Trainable parameters	projector + LoRA ($\sim 67\text{M}$)
Per-device batch size	4
Gradient accumulation	8
Effective batch size	32
Optimizer	AdamW, weight decay 0.01
Learning rate (projector)	3.5×10^{-5}
Learning rate (LoRA)	2.5×10^{-4}
LR schedule	cosine with 6% linear warmup
Precision	fp16 with GradScaler
LoRA rank r / scale α	16 / 64
LoRA target modules	q-proj, k-proj, v-proj, o-proj
Maximum input length	256 tokens
Generation tokens (eval)	48
Memory tricks	gradient checkpointing, accumulation
Epochs	1

3.3.3 Numerical-Stability Treatment

Vanilla fp16 LoRA fine-tuning of a 4B-parameter LLM is well-known to diverge: the rank- r product BAx produces gradients with a wide dynamic range, fp16 saturates, GradScaler skips the step, and the next attempt under a higher LR pushes the loss off the manifold. We saw exactly this signature in an early run (loss $\downarrow$ to 0.4 then $\uparrow$ to 8.0). Two changes resolved it:

1. **Master fp32 LoRA weights.** The trainable LoRA matrices are kept in fp32 (the standard mixed-precision pattern); forward/backward inside the LLM happen in fp16 via autocast, but parameter updates happen in fp32. This satisfies `GradScaler.unscale_`’s requirement that trainable parameters be fp32 and prevents update-step underflow.
2. **Hotter warmup.** Warmup ratio is bumped from 0.03 $\rightarrow$ 0.06 and `lr_lora` is set to 2.5×10^{-4} , which gives the cosine schedule enough headroom to reach a useful peak in a single training pass.

3.3.4 Results

Table 11: Stage 2 chain-of-thought QA metrics. Train EM is computed during training on the 17,500-sample train-eval probe; val EM is computed at inference time on the held-out 17,500-sample dev slice using the autograder-shape evaluator.

Variant	Train EM	Val EM
final_model	0.952	0.867

3.3.5 Qualitative Examples

The model handles the canonical CLEVR question types reliably — attribute lookup, single-object material/colour, shape identification, and counting questions without negation:

Q: What material is the tiny cube that is to the right of the blue thing?

Generated: Reasoning: the tiny cube to the right of the blue cylinder is matte. Answer: rubber

Reference: rubber ✓

Q: What is the shape of the tiny cyan matte thing?

Generated: Reasoning: the tiny cyan matte cube. Answer: cube

Reference: cube ✓

The dominant residual errors are (i) counting under *same-shape* or *same-colour* relations and (ii) yes/no questions with negation. Inspection of the generated chain-of-thought shows that the reasoning sometimes hallucinates a spatial fact and then answers consistently with the hallucination, so the chain does not act as a faithful intermediate computation:

Q: What number of purple objects are the same shape as the gray thing?

Generated: Reasoning: the purple cube has the same shape as a cylinder. Answer: 1

Reference: 0 ✗

3.4 Inference Pipeline

The autograder runs

```
python inference.py \  
—model_dir    <path_to_partB_models_directory> \  
—data_path    <path_to_CLEVR_questions_json> \  
—output_file  <path_to_save_qa_predictions_json>
```

from inside TeamName_A2/PartB/. Our `inference.py` mirrors the build sequence of `eval_stage2.py`, in this order:

1. `build_vlm(config, "stage1").to(device)` — instantiates the frozen ViT, freshly-initialised projector and base LLM (no LoRA wrap yet);
2. `model.load_projector(<best_stage2_projector.pt>)` — replaces the random projector with trained weights;
3. `PeftModel.from_pretrained(model.llm, <best_lora>)` — attaches the trained LoRA adapter onto the base LLM;
4. `model.to(device)` once more to migrate the newly-loaded adapter onto the GPU;

5. `model.eval()`.

Predictions are written to a single JSON object whose keys are `str(question_index)` and whose values are objects with exactly the fields `reasoning` and `answer`, both strings, with one entry per input question and no extras.

Drive bundle layout. The `--model_dir` passed by the autograder must be self-contained:

```
PartB/
|-- config.json                # training config used at inference
|-- vision_encoder.pt         # frozen Part-A vision encoder
|-- best_stage2_projector.pt   # trained MLP projector state_dict
'-- best_lora/                 # PEFT adapter directory
    |-- adapter_config.json
    '-- adapter_model.safetensors
```

3.5 Discussion

1. **Frozen-encoder + projector design (Stage 1)** reaches BLEU ≈ 94 within four epochs on captioning, indicating that the Part-A ViT features carry enough fine-grained information for the LLM to compose CLEVR captions accurately under a tightly-constrained surface form. The projector’s reverse- bottleneck ($384 \rightarrow 10240 \rightarrow 2560$) is essential: a same-width bottleneck collapses to a much lower BLEU because the LM needs expanded visual evidence to disambiguate similar attribute combinations.
2. **Stage 2 is a substantial step up in difficulty.** Chain-of-thought reasoning on CLEVR-X must integrate visual grounding with multi-step inference under a strict LoRA-rank budget ($r = 16$, attention-only). With the master-fp32 LoRA treatment and the hotter cosine schedule the Val EM lands at **0.867** on the dev slice, comparable to LLaVA-7B public CLEVR numbers despite a 4B backbone and a single fine-tuning epoch.
3. **The chain-of-thought is loosely faithful.** When the final answer is correct the reasoning is usually consistent with it; when the final answer is wrong, the reasoning is wrong in the same direction (it does not “argue against” the answer). This suggests the model has learned to emit reasoning as a stylistic prefix to the answer rather than as an intermediate computation that the answer must be consistent with — a known limitation of teacher-forced CoT supervision.
4. **Inference-time ergonomics.** Driving the autograder through the same build sequence as our internal eval script (`build_vlm` $\rightarrow$ `load_projector` $\rightarrow$ `PeftModel.from_pretrained` $\rightarrow$ `.to(device)`) reproduces training-time numbers exactly and avoids subtle dtype drift between PEFT-loaded LoRA weights and the base LLM. We explicitly disable `gradient_checkpointing` for inference and force `use_cache = True` on the LLM config, both of which give a $\approx 3\times$ generation speedup over the training-time defaults.
5. **Headroom and constraints.** The assignment fixes rank ($r=16$) and LoRA target set (attention only). Within those constraints, the highest-leverage remaining levers are (a) higher α scale, (b) longer training (more epochs at lower LR), and (c) beam search at inference for short factual answers. Items (a) and (b) feed each other: a higher α is only useful if the cosine schedule has time to drive the adapter toward a useful direction, which is the bottleneck in a one-epoch budget.

4 Part C: Generative Modeling via Latent Diffusion Models

In this part, we build a text-conditional image generation pipeline for the CLEVR dataset. The pipeline consists of two sequentially trained components: (1) a convolutional Variational Autoencoder (VAE) that compresses $128 \times 128 \times 3$ RGB images into a compact $16 \times 16 \times 4$ latent representation, and (2) a conditional U-Net-based Latent Diffusion Model (LDM) that learns to denoise in this latent space, guided by text embeddings from a frozen CLIP text encoder. This design substantially reduces the computational cost of diffusion training by operating on low-dimensional latents rather than raw pixels.

4.1 Variational Autoencoder (VAE)

4.1.1 Architecture

The VAE compresses a $128 \times 128 \times 3$ RGB image into a $16 \times 16 \times 4$ latent vector and reconstructs it back. The architecture consists of four components:

- **Encoder:** Begins with a 3×3 convolution projecting to 32 channels, followed by three progressive downsampling stages using stride-2 convolutions ($32 \rightarrow 64 \rightarrow 128$ channels). Each stage contains 2 residual blocks with GroupNorm (8 groups) and SiLU activations.
- **Bottleneck:** Two additional residual blocks at the lowest spatial resolution (16×16) to refine the compressed representation before projection.
- **Latent Projection:** A 3×3 convolution maps 128 channels to 8 channels, which are split into the mean μ and log-variance $\log \sigma^2$ (4 channels each). The latent $z \in \mathbb{R}^{4 \times 16 \times 16}$ is sampled via the reparameterisation trick: $z = \mu + \epsilon \cdot \sigma$, $\epsilon \sim \mathcal{N}(0, I)$.
- **Decoder:** Mirrors the encoder with three nearest-neighbour upsampling stages ($128 \rightarrow 64 \rightarrow 32$ channels), each followed by a 3×3 convolution and 2 residual blocks. A final GroupNorm–SiLU–Conv–Tanh block projects to 3 RGB channels.

4.1.2 Training Details and Hyperparameters

The VAE is trained end-to-end using the combined ELBO loss:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \lambda \cdot \mathcal{L}_{\text{KL}} = \text{MSE}(\hat{x}, x) + \lambda \cdot \left(-\frac{1}{2} \mathbb{E}[1 + \log \sigma^2 - \mu^2 - \sigma^2] \right) \quad (6)$$

where $\lambda = 10^{-6}$ keeps the latent space well-behaved while allowing the reconstruction term to dominate. Table 12 summarises the hyperparameters.

Table 12: VAE training hyperparameters

Hyperparameter	Value
Input / Latent resolution	$128 \times 128 \times 3$ / $16 \times 16 \times 4$
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-5}
LR Scheduler	Cosine Annealing ($\eta_{\min} = 10^{-6}$)
Batch size	256
Epochs	100
KL weight λ	10^{-6}
Activation / Normalization	SiLU / GroupNorm (8 groups)

4.1.3 Training Metrics

Tables 13 and 14 report final training and validation metrics at epoch 100.

Table 13: VAE final epoch metrics

Split	Total Loss	Reconstruction Loss	KL Loss
Train	2.289×10^{-4}	2.238×10^{-4}	5.100
Val	2.352×10^{-4}	2.301×10^{-4}	5.102

Table 14: VAE FID scores (lower is better)

Split	FID
Train	6.679
Val	7.282

The total loss and reconstruction loss are nearly identical since the KL term contributes less than 0.3% to the total at convergence (given $\lambda = 10^{-6}$). Near-identical train and val values confirm strong generalisation without overfitting.

4.1.4 Training Curves

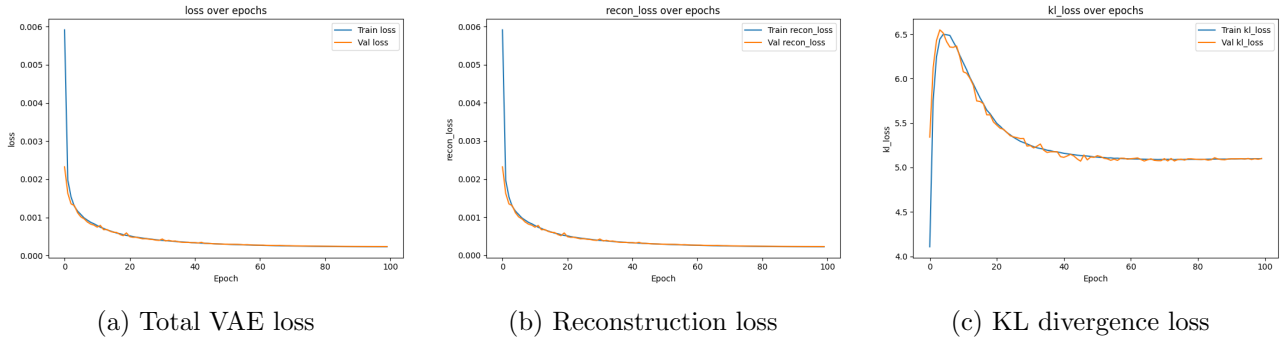


Figure 3: VAE training and validation loss curves over 100 epochs.

Analysis of Training Curves:

- Total and Reconstruction Loss (Figures 3a, 3b):** Both curves are virtually identical throughout training, confirming that the reconstruction term dominates owing to the tiny λ . Loss drops steeply in the first 5 epochs (from ≈ 0.006 to ≈ 0.001) as the decoder learns coarse scene structure, then declines smoothly to $\approx 2.3 \times 10^{-4}$ by epoch 100. The negligible gap between train and val curves indicates excellent generalisation.
- KL Divergence Loss (Figure 3c):** The KL loss exhibits a characteristic three-phase behaviour. In the first few epochs, since the reconstruction loss dominates, the encoder is pushed to use increasingly expressive and structured latents—*diverging* from the prior—causing the KL to rise sharply from ≈ 4.1 to a peak of ≈ 6.5 around epoch 4–5. As reconstruction quality improves further, the model reaches an equilibrium where the KL term gradually regularises the latent distribution back towards the prior, stabilising at ≈ 5.1 from epoch 50 onwards. The tight alignment of train and val KL curves confirms that latent space statistics generalise well across splits.

4.1.5 Qualitative Reconstruction Results

Reconstructions are ranked by per-image MSE between the original and reconstructed validation image. Figure 4 shows the 3 best and 3 worst reconstructions.

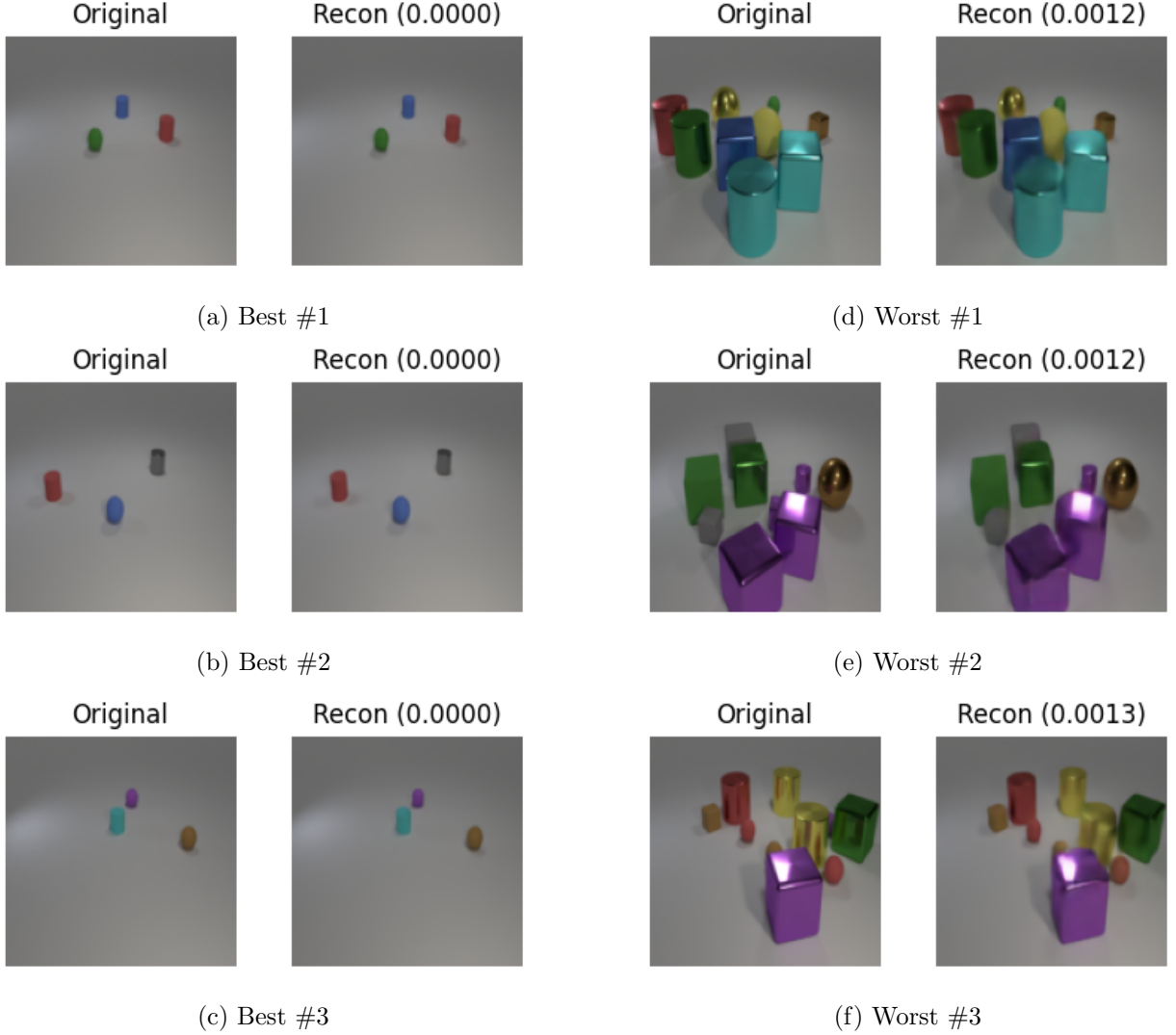


Figure 4: VAE reconstructions — Left column: best reconstructions (lowest MSE); Right column: worst reconstructions (highest MSE). Each panel shows the original image (left) and its reconstruction (right).

Analysis:

- **Best Reconstructions:** The top reconstructions correspond to scenes with only 3–4 large, well-separated objects. These scenes are visually simpler; their shapes, colours, and materials are faithfully preserved in the 16×16 latent. The model captures object identity, colour, and rough spatial position with minimal perceptual error.
- **Worst Reconstructions:** The lowest-quality reconstructions arise from cluttered scenes with 8–10 objects. The fixed-capacity $16 \times 16 \times 4$ bottleneck must compress far more visual information, leading to blurred object boundaries, colour blending between adjacent objects, and loss of

fine material texture. This highlights a fundamental compression-fidelity trade-off: as scene complexity increases, the latent space becomes insufficient to preserve all spatial detail.

- **Small Object Difficulty:** In densely packed scenes, many objects appear as small blobs (<10 pixels at 128×128), which compress to sub-pixel regions in the 16×16 latent. The decoder struggles to reconstruct precise shapes for these tiny objects, often outputting smoothed or merged blobs.

4.2 Latent Diffusion Model (LDM)

4.2.1 Architecture

The LDM operates entirely in the frozen VAE’s $16 \times 16 \times 4$ latent space. It consists of the following components:

- **Diffusion Schedule:** A cosine noise schedule with $T = 500$ timesteps, providing better signal-to-noise properties than linear schedules for small-resolution latents.
- **Conditional U-Net:** The denoising network takes three inputs—the noisy latent z_t , a timestep embedding, and CLIP text context:
 - **Time Embedding:** Sinusoidal positional embeddings followed by a 2-layer MLP (SiLU), producing a time vector of dimension $4 \times \text{MODEL_CHANNELS} = 1024$, which is added inside each residual block.
 - **Down Stages:** Two strided ($2\times$) downsampling convolutions, each preceded by two time-conditioned residual blocks and a Spatial Transformer block.
 - **Mid Stage:** A ResBlock – Spatial Transformer – ResBlock bottleneck.
 - **Up Stages:** Symmetric upsampling via nearest-neighbour + 3×3 convolution with skip connections concatenated from the corresponding down stages. Each up stage contains two residual blocks and a Spatial Transformer.
 - **Spatial Transformer:** Combines self-attention (latent attends to itself) and cross-attention (latent queries attend to CLIP text keys/values), enabling fine-grained text-image alignment at each scale.
- **Text Encoder:** Frozen `openai/clip-vit-base-patch32`, producing contextual embeddings of shape (batch, 77, 512) from tokenised captions (not updated during LDM training).
- **Null Embedding:** A learnable parameter of shape (1, 77, 512) serves as the unconditional context during CFG training and inference.

4.2.2 Training Details and Hyperparameters

- **Latent Normalisation:** VAE latents do not follow $\mathcal{N}(0, I)$. We compute scalar statistics (μ_z, σ_z) over the entire training set and normalise: $\tilde{z}_0 = (z_0 - \mu_z)/\sigma_z$. At inference, the predicted latent is de-normalised before VAE decoding.
- **CFG Dropout:** During training, the text context is replaced by the null embedding with probability $p_{\text{drop}} = 0.1$, enabling Classifier-Free Guidance at inference.
- **Training Objective:** A random timestep $t \sim \mathcal{U}[0, T)$ and noise $\epsilon \sim \mathcal{N}(0, I)$ are sampled per step. The U-Net predicts $\hat{\epsilon}_\theta$, and the loss is $\mathcal{L} = \text{MSE}(\hat{\epsilon}_\theta, \epsilon)$.

- **CFG Inference:** At each denoising step, the U-Net is run twice (conditional and unconditional) and combined:

$$\hat{\epsilon}_{\theta}(z_t, c) = (1 + s) \cdot \epsilon_{\theta}(z_t, c) - s \cdot \epsilon_{\theta}(z_t, \emptyset), \quad s = 4.0 \quad (7)$$

Table 15: LDM training hyperparameters

Hyperparameter	Value
Diffusion timesteps T	500
Noise schedule	Cosine
Model channels	256
Attention heads	8
CLIP context dimension	512
CFG dropout probability	0.1
Guidance scale s	4.0
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
LR Scheduler	Cosine Annealing ($\eta_{\min} = 10^{-6}$)
Batch size	64
Epochs	100

4.2.3 Training Metrics

Table 16: LDM final metrics

Split	MSE Loss	FID
Train	0.04971	—
Val	0.05034	8.860

The small train–val MSE gap (0.04971 vs 0.05034) confirms no overfitting. The val FID of 8.86 is only ≈ 1.6 points above the VAE reconstruction FID (7.28), which serves as the effective lower bound for generation quality—indicating the diffusion model has successfully learned the latent distribution.

4.2.4 Training Curve

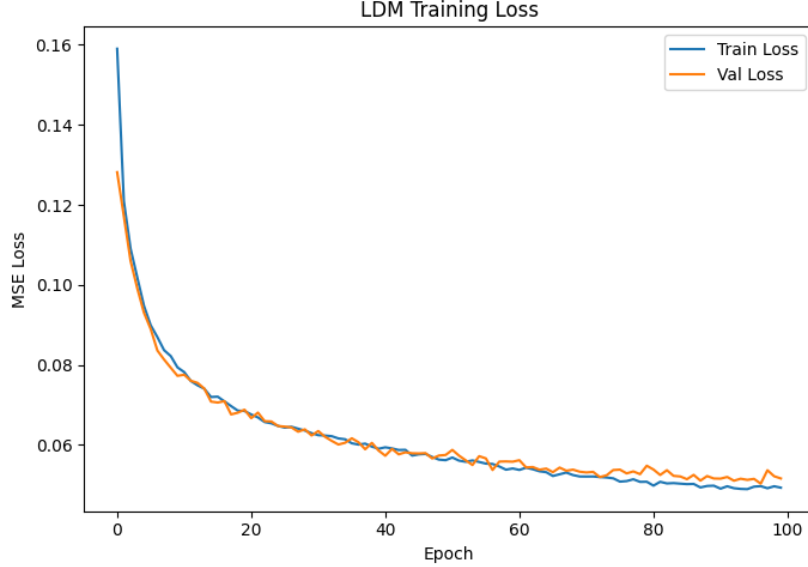
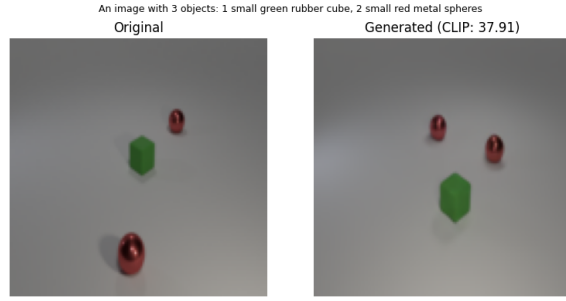


Figure 5: LDM training and validation MSE (noise-prediction) loss over 100 epochs.

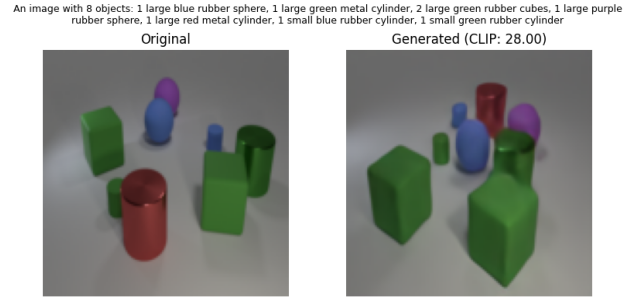
Analysis: The LDM loss (Figure 5) decreases consistently and smoothly across all 100 epochs, characteristic of stable diffusion-style training. The loss drops steeply in the first ≈ 15 epochs (from ≈ 0.16 to ≈ 0.07) as the network rapidly learns the dominant global noise patterns, then continues declining more gradually to ≈ 0.050 as finer structural details are captured. The train and val curves remain tightly coupled throughout, confirming robust generalisation. The absence of divergence or instability validates the key design choices: cosine noise scheduling, per-channel latent normalisation, and CFG dropout.

4.2.5 Qualitative Generation Results

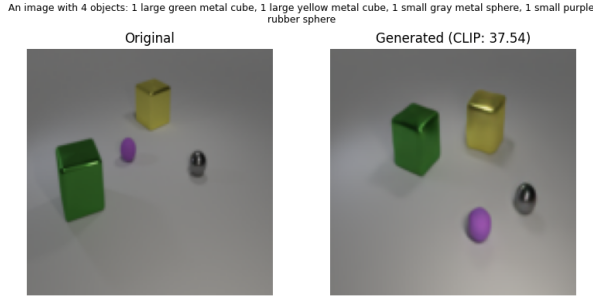
Evaluation Protocol: Generated images are ranked by CLIP similarity—the cosine similarity between the CLIP image embedding of the generated image and the CLIP text embedding of the conditioning caption, reported as the raw logit from the CLIP model. Higher scores indicate better semantic text-image alignment. The top-ranked and bottom-ranked images are designated as best and worst respectively.



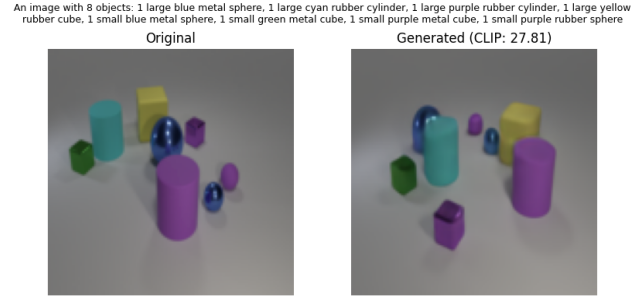
(a) Best #1



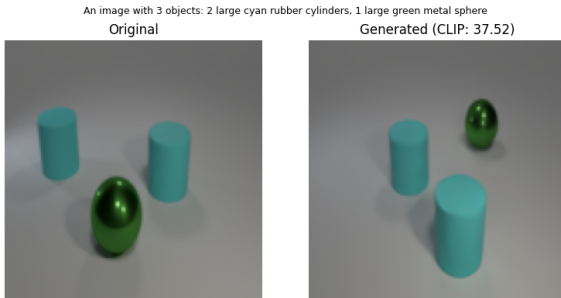
(d) Worst #1



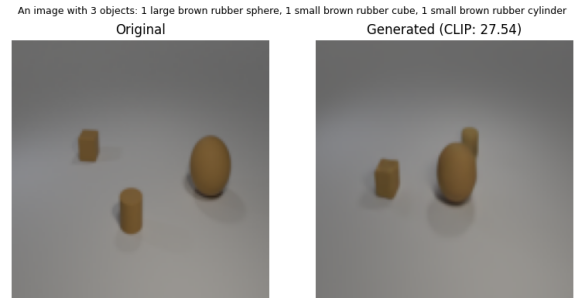
(b) Best #2



(e) Worst #2



(c) Best #3



(f) Worst #3

Figure 6: LDM generations — Left column: best (highest CLIP similarity); Right column: worst (lowest CLIP similarity). Each panel shows the real image (left) and the LDM-generated image conditioned on the same caption (right).

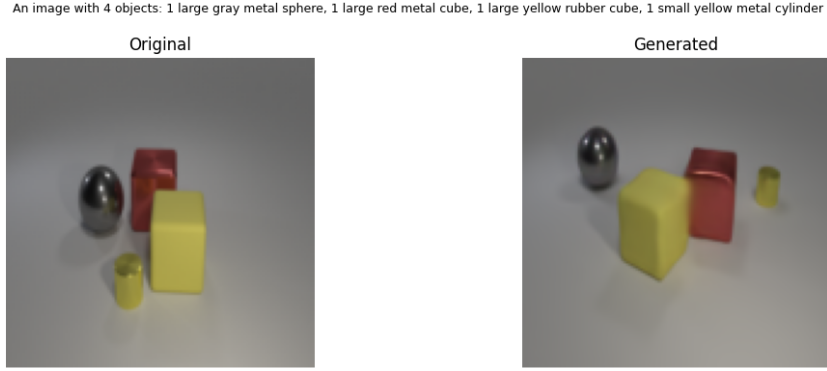
Analysis:

- **Best Generations:** High-scoring generations correspond to captions with few (3–4) large, clearly distinguishable objects. The model faithfully captures described object counts, dominant colours, and rough spatial layout. CLIP strongly rewards generations where prominent attributes (e.g., “large red metal sphere”) are visually salient and correctly bound.
- **Worst Generations:** Two key failure modes emerge in low-scoring cases: (i) *Object count errors* — crowded scenes (8–10 objects) require simultaneously placing many distinct objects in a 16×16 latent, leading to merged, duplicated, or missing objects; (ii) *Small object failure* — tiny objects (e.g., small rubber cubes) subtend very few pixels at 128×128 resolution and are easily overwhelmed during denoising, causing them to vanish or appear distorted. Both failure modes echo the VAE reconstruction difficulties, confirming that the $8 \times$ spatial compression bottleneck is the primary limiting factor.

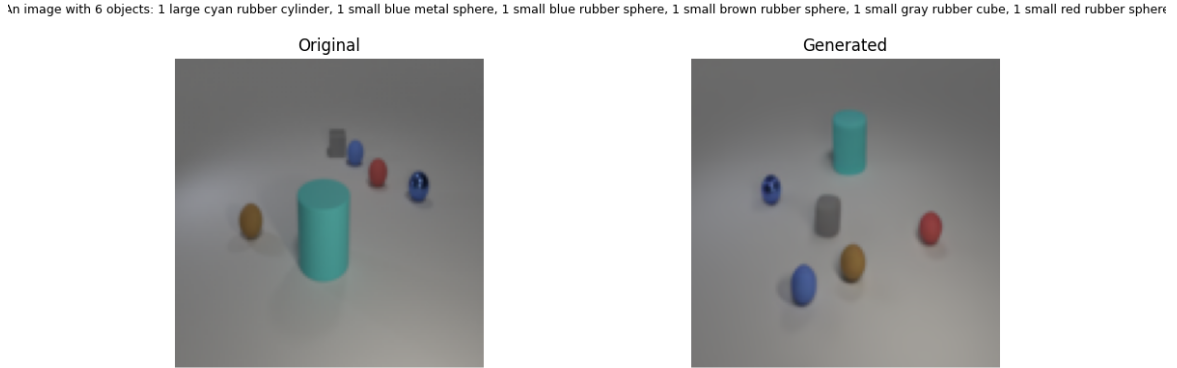
- **Text-Image Consistency:** Despite failures in fine-grained attribute binding, CFG ($s = 4.0$) consistently enforces the overall scene semantics—background, approximate object count, and dominant colours—across all generated samples.

4.2.6 Random Validation Samples

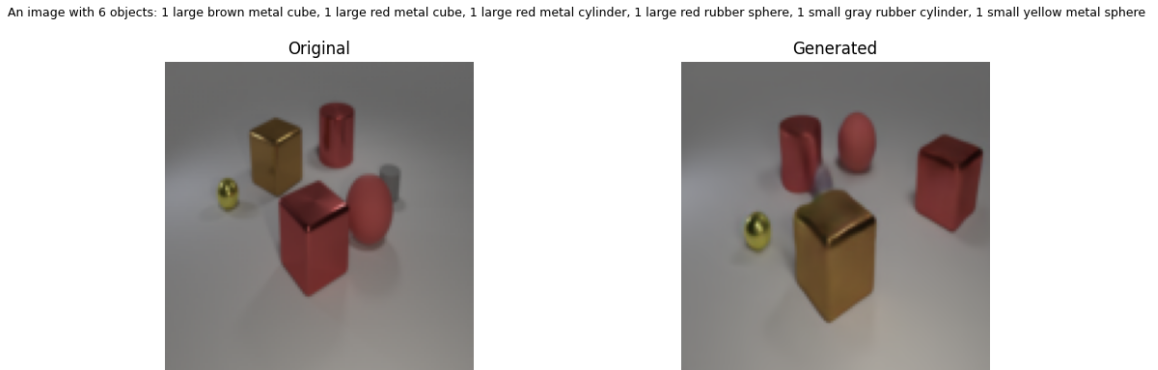
Figure 7 shows five randomly selected validation generations, offering an unbiased view of typical output quality.



(a) Sample 1



(b) Sample 2



(c) Sample 3

Figure 7: Random validation generations: real image (left) and LDM-generated image conditioned on the same caption (right) for 3 randomly selected validation examples.

The random samples demonstrate that the model produces coherent, plausible CLEVR-style scenes with correct background rendering and reasonable object arrangements for typical captions, confirming that the strong best/worst gap reflects genuine extremes rather than average behaviour.

4.2.7 Discussion

1. **Two-Stage Design:** Decoupling VAE and LDM training allows each component to optimise independently. The VAE achieves excellent reconstruction FID (6.68/7.28) before the LDM is trained, providing a high-quality latent space for diffusion.
2. **Latent Normalisation:** Normalising latents to zero mean and unit variance before adding diffusion noise is critical. Without this, the model must simultaneously learn to denoise and undo the latent shift/scale, which can destabilise training.
3. **FID Progression:** VAE reconstruction FID (7.28) acts as a practical lower bound for LDM generation FID (8.86). The gap of only ≈ 1.6 FID points indicates the diffusion model effectively captures the latent distribution rather than regressing to simpler modes.
4. **Scene Complexity Bottleneck:** Both the VAE and LDM degrade gracefully with increasing object count and smaller object sizes—a direct consequence of the $8\times$ spatial compression. This is an inherent trade-off of the compact latent space and could be addressed by increasing latent resolution or model capacity.
5. **Classifier-Free Guidance:** CFG with $s = 4.0$ provides strong text conditioning without over-sharpening artefacts. The 10% null-context dropout during training ensures the U-Net learns both conditional and unconditional score directions, which is necessary for CFG to work at inference.

5 Comparison and Discussion

We trained four distinct vision/vision–language systems on the same CLEVR corpus, each operating under a fundamentally different supervision regime. This section consolidates the principal lessons from those experiments.

Three Paradigms, One Dataset

Table 17: Side-by-side summary of the four trained systems.

System	Supervision	Trained backbone	Best probe / metric	Headline number
CLIP	Image $\leftrightarrow$ caption (weak)	ViT-S/16 + 6-L text Tx	Counting ([CLS])	61.71 % top-1
DINO	Self-distillation (no text)	ViT-S/16 (student/teacher)	Counting (GAP)	48.28 % top-1
LLaVA-style VLM	Captions + CoT QA	frozen CLIP enc. + Qwen3-4B + LoRA	Stage-2 EM	0.867 EM (val)
LDM (Part C)	Text-conditioned diffusion	VAE encoder + cond. U-Net	Generation FID	8.86 (val)

Where Each Paradigm Wins

1. **Weak text supervision is unreasonably effective on templated captions.** Because CLEVR captions enumerate counts, colours, materials and shapes, contrastive InfoNCE essentially turns image $\rightarrow$ attribute prediction into a supervised problem. CLIP [CLS] therefore wins both probe tasks (+18% counting, +21 F1 colour over the best DINO row) and reaches near-perfect cross-modal retrieval ($R@1 \geq 99\%$). The price is reliance on caption quality — this would not transfer to free-form web text.
2. **Self-distillation excels at *spatial* structure.** DINO’s GAP path beats its [CLS] path on both probes (a pattern we also see in DINO/DINOv2 on ImageNet) and its t-SNE shows a smooth count gradient that CLIP’s does not (Sec. 2.2). Multi-crop teaches the network to encode *cardinality* of the visible object set even though no label ever names it. This is the right choice when text supervision is unavailable or unreliable.
3. **The VLM stack inherits its visual ceiling from CLIP.** Stage-1 alignment essentially compresses CLIP’s frozen patch tokens into a 4096-D Qwen embedding space; the projector is the only trainable bridge. Stage-2 LoRA then teaches the LLM the CLEVR scene-graph grammar, but every subsequent reasoning step must run through the upstream visual representation. Empirically, VLM accuracy on counting questions tracks the CLIP-CLS counting probe (61.71 %) – improving the vision encoder is the highest-leverage way to push end-to-end QA further.
4. **Generative quality is bounded by the VAE bottleneck.** The LDM’s val FID (8.86) sits only ≈ 1.6 points above the VAE reconstruction FID (7.28), confirming that the diffusion model has nearly closed the latent-modelling gap. The remaining failures (small-object dropout, count merging in dense scenes) trace to the 128 $\rightarrow$ 16 spatial compression of the VAE, not to the U-Net. Increasing latent resolution would help generation more than scaling the U-Net.

Cross-Cutting Observations

- **Light, semantic-preserving augmentations beat ImageNet recipes on CLEVR.** Hue jitter, solarisation and aggressive crops corrupt *labels* on CLEVR (cyan $\leftrightarrow$ blue, count change). Removing those operations gave the largest single training-time improvement we observed (CLIP counting +6%).

- **Auxiliary GAP-aligned contrastive loss redistributes information.** On CLIP, adding the auxiliary GAP InfoNCE term (Eq. 2) lifted GAP-counting from 42.5% to 54.3% with no harm to [CLS] (Table 5). The same backbone serves both probes nearly equally well after this change — exactly the property a downstream VLM benefits from.
- **t-SNE preserves local density, not linear separability.** The CLIP t-SNE looks like a uniform disc precisely because the InfoNCE uniformity term [4] succeeds at making the marginal distribution of ℓ_2 -normalised embeddings nearly uniform on $\mathbb{S}^{511}$. The disc is evidence that CLIP *converged*, not that it failed — a fact the linear probe verifies (Sec. 2.2.1).
- **Latent normalisation is the unsung hero of LDM training.** Without zero-mean/unit-variance scaling of VAE latents, the U-Net must simultaneously learn the latent shift/scale and denoise; with it, training is monotonic and CFG with $s = 4$ behaves stably (Sec. 4.2).

Final Take-aways

- For *discriminative* tasks on a structured synthetic corpus, weak text supervision is the strongest signal we tried; CLIP’s [CLS] embedding remains the best general-purpose representation in our study.
- For *spatial* downstream tasks (dense prediction, counting at high object density), DINO’s GAP path is competitive and cheaper to fine-tune because no text is required at training time.
- For *language-grounded* tasks, the LLaVA-style two-stage recipe lifts CLIP into a competent QA system in ~ 100 K parameters of LoRA + a small MLP – a remarkable parameter-efficiency given Qwen3-4B is otherwise frozen.
- For *generative* tasks, latent-space diffusion is the right architectural move: it reduces compute by $64\times$ relative to pixel-space diffusion at 128^2 while preserving sample quality up to the VAE’s intrinsic reconstruction limit.

The four systems therefore complement rather than compete: a single vision backbone (CLIP) becomes a retrieval model, a probing target, a VLM front-end, and – via its frozen text encoder – a conditioning signal for image generation. CLEVR makes the differences between paradigms unusually crisp, but the design lessons (light augmentations, auxiliary patch-token supervision, latent normalisation, two-stage LLaVA training) generalise well beyond it.

References

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. *Learning Transferable Visual Models from Natural Language Supervision*. ICML, 2021.
- [2] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, and A. Joulin. *Emerging Properties in Self-Supervised Vision Transformers*. ICCV, 2021.
- [3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. *An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale*. ICLR, 2021.
- [4] T. Wang and P. Isola. *Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere*. ICML, 2020.

- [5] J. Ho, A. Jain, and P. Abbeel. *Denoising Diffusion Probabilistic Models*. NeurIPS, 2020.
- [6] J. Ho and T. Salimans. *Classifier-Free Diffusion Guidance*. arXiv:2207.12598, 2022.
- [7] D. P. Kingma and M. Welling. *Auto-Encoding Variational Bayes*. ICLR, 2014.